

Benjamin Lee

Capstone Write-Up

The Tool

The tool I'm using is called Relevance AI. It is a agent building platform that enables fairly low-code creation of autonomous AI workflows, LLM reasoning, and tool usage. It is an agentic enablement tool because it allows the creation of agents that can interpret, act, run code, provide analysis, and operate with autonomy once directed in natural language. We have not used it in this course yet, and it thus qualifies as new.

I chose this tool because I was looking for something that had robust tool selection and orchestration, particularly in the area of analytics, as well as a low-code environment. I am using the free tier, which played into my decision as well. The free tier allows for a large token allowance and most the functionality on the platform.

The Build

I built an assistant data analyst chatbot for my project. It currently has only Newport News Arrest data from the past 5-years in its knowledge base, but in theory this could be broadened to other open data assets. It is designed to answer natural language questions about the data like:

1. What are the most common arrest charges?
2. How many arrests happen each year?
3. Which locations have the highest number of arrests?
4. What is the distribution of arrest types?

The intended audience for this particular agent is the public who are looking to access addition information from the large datasets that are made available via the City's Open Data Portal. This includes civic researchers, journalists, community stakeholders, and even potentially city staff who can get all of the required information from the open data asset.

Link to working chat portal: https://app.relevanceai.com/agents/bcbe5a/29680334-ee07-4357-93f5-6d467707432a/c6f3b853-d023-4a70-b5e9-4acbf64969a4/embed-chat?hide_tool_steps=true&hide_file_uploads=true&hide_conversation_list=true&bubble_style=agent&primary_color=%23685FFF&bubble_icon=pd%2Fchat&input_placeholder_text=Type+your+message...&hide_logo=false&hide_description=true (Please do not use until after the demo, I have limited tokens)

Agent Card

Purpose:

This agent provides accurate descriptive analytics on the last five years of Newport News arrest data to support the public and civic researchers.

Role:

It is a data analyst assistant optimized for interpreting analytical questions, mapping them to the correct dataset fields via a semantics layer, and generating statistically correct, reproducible outputs using Python.

Inputs:

1. Can Access:
 - a. CSV file containing 5-years of arrest data in the City of Newport News.
 - b. All fields defined in the dataset's semantics layer.
 - c. Python code execution for statistical computations, operations, and data transformations.
2. Can't Access:
 - a. Arrest, investigation, and/or case notes.
 - b. Arrest data outside the given timeframe.
 - c. Arrest outcomes or PD decision making process or operational decisions.

Task:

1. Read the user's analytics question.
2. Use the semantics layer to identify the correct dataset fields; never guess or invent columns.
3. Determine the appropriate tool:
 - a. If the question involves counting, frequency, "top N," "most common," or "distribution," select the Python execution tool and run `pandas value_counts`.
 - b. If the question requires cross-field interpretation or multi-step analysis, select the insights tool.
4. Load the arrest dataset from the file.
5. Execute the correct analytical workflow:
 - a. For counting/frequency: run optimized pandas operations.
 - b. For multi-step logic: run code or insights as appropriate.
6. Convert the raw output into clear, business-friendly language describing the insight.
7. If the query is ambiguous, ask clarifying questions before running any analysis.
8. If the data needed for the user's question does not exist, state this directly and provide possible alternatives.
9. If any tool error occurs, inform the user and escalate to Benjamin Lee.

Constraints:

1. Never fabricate column names, values, categories, or statistics.

2. Never infer or imply causation from the arrest dataset.
3. Never make statements about guilt, innocence, or predictive risk.
4. Never provide personally identifiable information beyond what the dataset explicitly contains
5. Never analyze demographic attributes beyond descriptive statistics
6. Never bypass tool-selection rules; all counting must use pandas value_counts via Python.
7. Never proceed with analysis if the dataset cannot be loaded.
8. Never perform analyses outside the scope of the dataset's time range.

Output Format:

1. Markdown-formatted response
2. First: a plain-language interpretation of the result (2–4 sentences)
3. Then: a cleanly formatted table or list of computed values, if relevant
4. Use professional, neutral, civic-analytics tone
5. Cite field names exactly as they appear in the semantics layer
6. If unable to answer: provide a short explanation and an alternative analysis suggestion

Escalation Trigger:

Escalate immediately and output: “This request requires human review. Reason: [insert reason].”

If any of the following occur:

1. The agent cannot match the user's terms to a valid field in the semantics layer
2. The dataset fails to load or appears corrupted
3. The user asks for predictive, causal, or policy recommendations
4. The question involves identifying individuals or producing sensitive personal details
5. The question requires information not present in the dataset

Success Metric:

At least 90% of analytical outputs within a random sample meet all four evaluation criteria: correct field mapping, correct computation, clear explanation, and no hallucinated fields.

Evaluation

Correctness: Score: 10/10

The agent consistently used the appropriate dataset fields. It did not hallucinate fields during testing and correctly mapped user language to the proper columns. It used the correct tool (Python execution with pandas) for all frequency-based questions. Minor limitations remain in handling ambiguous queries that reference vague requests.

Completeness: Score: 8/10

In each test run, the agent delivered structured results followed by an interpretation suitable for civic analytics reporting. When information was unavailable (e.g., demographic fields not

included), the agent correctly stated limitations or simply returned zeros for the questions statistics. This failure to note that something doesn't exist is a nuanced failure and one that I haven't been able to fix yet.

Safety: Score: 10/10

The agent adhered to safety constraints. It doesn't draw causal claims from arrest data or fabricated values. Escalation triggers on failures.

Fit: Score: 8/10

The agent's responses (explanations and outputs) are appropriate for the audience. Statistical summaries were clear and not highly technical. The agent did not ask great clarifying questions, which is a concern for usefulness in public use. Overall though, it aligns well with the real-world usecase.

Test Runs:

1.

The first screenshot shows a user query: "What is the most numerous arrest charge?" The agent responds with a summary and a list of top charges. The second screenshot shows a user query: "Let's filter to the last 30 days." The agent responds with a summary of the filtered data and a list of top charges in the last 30 days. The third screenshot shows a user query: "Let's filter to the last 30 days." The agent responds with a summary of the filtered data and a list of top charges in the last 30 days.

Timeline +

Triggered by Benjamin Lee

Most Common Arrest Charge Inquiry 4 hours ago

What is the most numerous arrest charge?

1 step performed in the background >

Data Analyst Agent provided an update:

Summary

- I used the dataset column named "Charge" to answer this (the file contains a Charge column).
- Most numerous arrest charge: ASSAULT: ON FAMILY MEMBER — 5,681 records.
- This represents about 11.5% of the non-null Charge entries (5,681 of 49,309).

Top charges (from the Charge column, top 10)

1. ASSAULT: ON FAMILY MEMBER — 5,681
2. FAILURE TO APPEAR: ON MISDEMEANOR OFFENSE — 3,284
3. ASSAULT: (MISDEMEANOR) — 2,566
4. DRUGS: POSSESS SCH I OR II — 2,254
5. SHOPLIFT/ALTER PRICE: LARCENY <\$200, 1ST OFF — 1,759
6. DWI: 1ST OFF, BAC .08-.14% — 1,638
7. TRESPASS: AFTER BEING FORBIDDEN TO DO SO — 1,625
8. CONCEALED WEAPON: CARRY — 1,516
9. PROFANE SWEARING OR INTOXICATION IN PUBLIC — 1,075
10. PROBATION: VIOLATION ON FELONY OFFENSE — 965

Notes and next steps

- I selected the Charge column because it contains the arrest charge text; there were 49,309 non-null values in that column and 693 unique charge descriptions.
- If you'd like, I can:
 - Return a full table of charges with counts (top N or complete list).
 - Break this down by Date, Precinct, Race, Sex, or other fields.
 - Normalize similar charge text (e.g., merge slightly different strings that refer to the same underlying charge) before counting. Which follow-up would you like?

Timeline +

Benjamin Lee replied with the following:

Let's filter to the last 30 days.

2 steps performed in the background >

Considering date parsing details...

Data Analyst Agent provided an update:

I filtered the dataset by the Date column to the last 30 days (period: 2026-05-17 through 2026-06-16) and counted charges in that window.

Key results

- Date field used: Date
- Period filtered: 2026-05-17 → 2026-06-16 (last 30 days)
- Rows with parsed Date in period: 593
- Non-null Charge entries in period: 591
- Most numerous charge (last 30 days): ASSAULT: ON FAMILY MEMBER — 71 occurrences (=12.0% of non-null charges in the period)

Timeline +

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Timeline +

Top charges in the last 30 days (top 10)

1. ASSAULT: ON FAMILY MEMBER — 71
2. FAILURE TO APPEAR: ON MISDEMEANOR OFFENSE — 39
3. INTOXICATION IN PUBLIC; ... TRANSPORTATION OF PUBLIC INEBRIATES TO DETOXIFICATION CENTER — 33
4. DRUGS: POSSESS SCH I OR II — 33
5. ASSAULT: (MISDEMEANOR) — 28
6. CONCEALED WEAPON: CARRY — 25
7. SHOPLIFT/ALTER PRICE: LARCENY <\$200, 1ST OFF — 20
8. DWI: 1ST OFF, BAC .08-.14% — 17
9. ABDUCTION: BY FORCE, INTIMIDATION OR DECEPTION — 16
10. TRESPASS: AFTER BEING FORBIDDEN TO DO SO — 15

Notes and options

- I parsed Date with pandas; rows with unparseable dates were excluded from the 593 count.
- If you want: I can return the full counts table (top N or complete list), break these counts down by Precinct / Race / Sex / Day, or normalize similar charge text before counting (recommended if you want merged variants). Which would you like next?

Leave comment for Data Analyst Agent

2.

Timeline +

B Benjamin Lee replied with the following: just now

Can you tell me the top 5 charges by month over the last 5 years?

1 step performed in the background > just now

Summarizing charges data

I'm mapping results from June 2021 to June 2026, covering 61 months. If I present 37,814 rows with 637 unique charges, it might be overwhelming. The user asked for the top 5 charges by month over the last 5 years, so I should show that directly while considering a summary alongside. It'll make sense to present key patterns, listing the top 5 charges for each month and maybe grouping by year for clarity. A concise output should work well!

Constructing user summary

Following the developer's guidelines, I need to reference relevant fields like Date and Charge. I should mention that months without records will show empty. It's also essential to offer next steps, such as providing a CSV or a breakdown by precinct. My final message will include a brief summary of the timeframe from June 2021 to June 2026, covering 61 months. I can present a code block with top 5 charges listed by month, and then offer an export option for CSV.

Creating month entries

I'm going to format the month entries in a compact way, like "2026-06: ASSAULT: ON FAMILY MEMBER — 40; FAILURE TO APPEAR...". It's important to keep the ordering correct, ensuring the months go from June 2021 to June 2026 in ascending order. This will make the results clear and easy to follow. Now, I'll put together the final answer with this format.

Leave comment for Data Analyst Agent

Timeline +

Data Analyst Agent provided an update: just now

Short summary

- Data used: Date (for month) and Charge (charge text).
- Period covered: 2021-06 through 2026-06 (last 5 years from today).
- Rows in that 5-year period (with parsed Date and non-null Charge): 37,814.
- I computed the top 5 charges (by count) for each month.

Top 5 charges by month (month: charge — count)

- 2026-06: ASSAULT: ON FAMILY MEMBER — 40; FAILURE TO APPEAR: ON MISDEMEANOR OFFENSE — 23; DRUGS: POSSESS SCH I OR II — 20; ASSAULT: (MISDEMEANOR) — 15; CONCEALED WEAPON: CARRY — 12
- 2026-05: ASSAULT: ON FAMILY MEMBER — 60; INTOXICATION IN PUBLIC; PENALTY; TRANSPORTATION OF PUBLIC INEBRIATES TO DETOXIFICATION CENTER — 40; DRUGS: POSSESS SCH I OR II — 33; ASSAULT: (MISDEMEANOR) — 31; FAILURE TO APPEAR: ON MISDEMEANOR OFFENSE — 28
- 2026-04: ASSAULT: ON FAMILY MEMBER — 76; FAILURE TO APPEAR: ON MISDEMEANOR OFFENSE — 57; ASSAULT: (MISDEMEANOR) — 37; DRUGS: POSSESS SCH I OR II — 36; SHOPLIFT/ALTER PRICE: LARCENY <\$200, 1ST OFF — 35
- 2026-03: ASSAULT: ON FAMILY MEMBER — 77; FAILURE TO APPEAR: ON MISDEMEANOR OFFENSE — 57; DRUGS: POSSESS SCH I OR II — 43; CONCEALED WEAPON: CARRY — 37; INTOXICATION IN PUBLIC; PENALTY; TRANSPORTATION OF PUBLIC INEBRIATES TO DETOXIFICATION CENTER — 35

Leave comment for Data Analyst Agent

Timeline +

B Benjamin Lee replied with the following: just now

Can you put these into a table for me?

Creating a Markdown table

I need to respond using markdown since the developer message requires it. The user asked to put the previously computed "top 5 charges by month" into a table. Given that there are 61 months, it's important to present this cleanly. I'll offer options for a full table or a downloadable CSV/Excel, but since they want a direct table, I'll provide it within markdown. I can create a concise version with columns like Month, Rank1 Charge, Rank2, through Rank5. That should be acceptable!

Preparing the Markdown Table

I need to make sure not to invent any values, so I'll use the previously computed data from the "top5_by_month" dictionary. I can create the markdown table with columns for Month and the top five charges. There will be 61 rows from June 2021 to June 2026. Although it's a long list, the user requested it. I'll also mention the option for a CSV or Excel download and ask if they prefer a different order. Let's get the rows compiled!

Crafting the Markdown Table

The user didn't specify if they want the table in ascending or descending order, but previously the assistant suggested the newest items at the top. I think I'll ask if they have a preference. They did ask for a table, so I'll likely present it with the latest months first.

I'll create a markdown table with a Month column and five columns for Rank 1-5 in chronological order from June 2021 to June 2026. It'll be a large table, with 61 rows, but I'll format it carefully, ensuring each entry is correct, including filling missing cells as needed. Let's get started!

Creating the Markdown Table

I'm going to use content from the previous assistant message to ensure the counts match. Although the user didn't specify, I'll maintain the descending order starting from June 2026 to June 2021, which aligns with the earlier output.

I need to carefully prepare the 61 rows to ensure there are no mistakes. I'll copy from that earlier message, starting with the top entries for each month. I'll make sure to double-check the counts while filling in the table correctly. Let's finalize this table!

Constructing the Markdown Table

I'm making sure to accurately check the counts for each month's data to ensure everything is in order. For each month, I need to confirm that the entries match up correctly, especially when counts tie.

I'll create a markdown table with six columns: Month, Rank 1, Rank 2, Rank 3, Rank 4, and Rank 5. Given the table's length, I'll include a note offering a CSV download for convenience and ask if the user prefers a different format. Finally, I'll follow the markdown rules, keeping everything organized and clear!

Leave comment for Data Analyst Agent

Timeline +

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Leave comment for Data Analyst Agent

Below is the monthly top-5 table (Month | Rank 1 → Rank 5). Each cell shows "Charge — count". I kept the same month range used earlier: 2021-06 through 2026-06.

Month	1	2	3	4	5
2026-06	ASSAULT: ON FAMILY MEMBER — 40	FAILURE TO APPEAR: ON MISDEMEANOR OFFENSE — 23	DRUGS: POSSESS SCH I OR II — 20	ASSAULT: (MISDEMEANOR) — 15	CONCEALED WEAPON: CARR — 12
2026-05	ASSAULT: ON FAMILY MEMBER — 60	INTOXICATION IN PUBLIC; ... TRANSPORTATION OF PUBLIC INEBRIATES TO DETOXIFICATION CENTER — 40	DRUGS: POSSESS SCH I OR II — 33	ASSAULT: (MISDEMEANOR) — 31	FAILURE TO APPEAR: ON MISDEMEANOR OFFENSE — 28
2026-04	ASSAULT: ON FAMILY MEMBER — 76	FAILURE TO APPEAR: ON MISDEMEANOR OFFENSE — 57	ASSAULT: (MISDEMEANOR) — 37	DRUGS: POSSESS SCH I OR II — 36	SHOPLIFT/ALTE PRICE: LARCEN <\$200, 1ST OFF 35
2026-03	ASSAULT: ON FAMILY MEMBER — 77	FAILURE TO APPEAR: ON MISDEMEANOR OFFENSE — 57	DRUGS: POSSESS SCH I OR II — 43	CONCEALED WEAPON: CARRY — 37	INTOXICATION I PUBLIC; ... TRANSPORTATI OF PUBLIC

Note: I did not paste the

full tables, they are huge, but complete.

3.

Triggered by Benjamin Lee

Arrestee Generation Breakdown Request

2 minutes ago

Can you please break down arrestees by generation and by year. Generations are Silent Generation (Ages 81-98), Baby Boomers (Ages 62-80), Generation X (Ages 46-61), Millennials (Generation Y) (Ages 30-45), Generation Z (Ages 14-29).

Good

Bad

3 steps performed in the background just now

Summarizing generation counts

I've got the sensible yearly counts from the Date column for 2020-2026. I need to present these results in markdown, mapping the generations and methods used, mentioning the fields "Age" and "Date." A breakdown table per year is essential, along with overall counts. Per the developer rules, I'll validate field references and note any counts for "Under 14" or "Over 98". Finally, I'll compute the total arrests, which sums to 49,827.

Confirming totals and presentation

I need to confirm totals, and while I could run Python to compute them, it's fine to just sum them up directly. I should be precise since some rows might have missing ages labeled as 'Unknown', but the counts_by_generation dict is full. Maybe I'll run a quick Python calculation, but I don't want to make unnecessary calls. It's fine to sum. I'll present a per-year table with the counts and overall totals across all years for each generation.

Preparing notes and summary

Leave comment for Data Analyst Agent

Preparing notes and summary

I'll provide notes that I used the 'Age' column for ages and the 'Date' column for the arrest year extraction. I'll explain the generation definitions for the user and mention that any ages outside defined ranges were labeled as 'Under 14' or 'Over 98', but none are present. If the user wants a CSV, different groupings, adjusted generation ranges, or to exclude missing ages, I'll ask. I'll follow developer instructions and format everything in markdown, including a concise summary and ASCII tables for overall counts and year-by-generation. Millennials and Gen Z together make up a majority, with percentages around 43.4% and 35.1%, respectively.

Computing percentages and insights

I'll quickly compute approximate percentages based on the total of 49,827.

Millennials make up about 43.4%, Gen Z around 35.1%, Gen X is 17.1%, Boomers 4.4%, and Silents at 0.07%. I should mention that the years covered are 2020-2026, and the counts per year seem relatively stable, with Millennials being the largest group each year. I might offer to export a CSV or produce a chart and ask if the user wants percentages per year. Since we've broken down arrestees by generation and year, I'll prepare the final message in markdown with tables and explanations.

Data Analyst Agent provided an update: just now

Summary

- I used the dataset fields Age and Date (both detected in the CSV) to assign each arrestee to the generation ranges you specified and to extract the arrest year.
- Generation definitions used (per your message):
 - Silent Generation: Ages 81-98
 - Baby Boomers: Ages 62-80

Leave comment for Data Analyst Agent

Summary

- I used the dataset fields Age and Date (both detected in the CSV) to assign each arrestee to the generation ranges you specified and to extract the arrest year.
- Generation definitions used (per your message):
 - Silent Generation: Ages 81-98
 - Baby Boomers: Ages 62-80
 - Generation X: Ages 46-61
 - Millennials: Ages 30-45
 - Generation Z: Ages 14-29
- Rows with ages outside those ranges were labeled Under 14 / Over 98 (none of those groups appear in the main counts below).
- Time coverage in the data: arrests dated 2020 through 2026.

Overall counts (by generation)

- Total arrestees counted: 49,827

Generation	Count	% of total
Millennial (30-45)	21,611	43.4%
Generation Z (14-29)	17,467	35.1%
Generation X (46-61)	8,526	17.1%

Quick observations

- Millennials are the largest group across the dataset (~43%), followed by Generation Z (~35%). Generation X is the third-largest (~17%).

Generation X (46-61)

8,526

17.1%

Baby Boomer (62-80)

2,187

4.4%

Silent (81-98)

36

0.07%

Breakdown by year (counts)

Year	Silent (81-98)	Baby Boomer (62-80)	Generation X (46-61)	Generation Z (14-29)	Millennial (30-45)
2020	0	255	1,370	3,648	3,249
2021	0	242	1,201	2,443	2,892
2022	5	318	1,194	2,484	3,125
2023	12	340	1,400	2,720	3,673
2024	7	459	1,332	2,578	3,593
2025	8	398	1,433	2,439	3,513
2026	4	175	596	1,155	1,566

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- Millennials are the largest group across the dataset (~43%), followed by Generation Z (~35%).

Quick observations

- Millennials are the largest group across the dataset (~43%), followed by Generation Z (~35%). Generation X is the third-largest (~17%).
- Year-to-year counts show the bulk of arrests are in 2020-2025, with similar generation proportions each year; 2026 currently has fewer records (partial year or incomplete data).

Method notes and data decisions

- Fields used: Age (for generation assignment) and Date (for extracting arrest year). I prioritized the explicit 'Date' column for year extraction.
- If you want different generation cutoffs, or to compute generations from date-of-birth instead of Age (if DOB exists or if you prefer age-at-arrest calculated from DOB), I can re-run with that logic.
- If you want the result exported as CSV or a visualization (bar chart by year and generation, stacked bar, etc.), tell me which format and I'll produce it.

Next steps / Questions

- Is the 'Date' column the intended arrest date to use for year? (I used it.)
- Do you want percentages by year (e.g., share of arrests that year by generation) in addition to raw counts?
- Would you like a downloadable CSV or charts to visualize the breakdown?

If anything looks off or you want adjustments, tell me which changes you'd like and I will re-run the analysis.

Risk and Governance

1. Prompt Injection:

- Prompt injection could be a concern for my agent. However, given that my agent does not have any non-public data in its knowledge base, I am less concerned about prompt injection than I would be with an agent which had access to sensitive information. This is the chief mitigation in my agent. A concern about

prompt injection is that a user may convince the model to say something that would damage the reputation of the City. To mitigate this I have noted that the agent must avoid making causal claims, particularly around demographic data.

2. Tool Abuse Risks:

- a. The only tools that this agent has access to is the python tool and the relevance ai insights tool. Given this, risks are fairly minimal. However, there is a possibility that users may try and overload the agent with complex requests, costing the City a large amount of money. To stop this I have capped the number of attempts the agent can make and the number of credits/tokens it can use on a task.

3. Data Leakage

- a. Given that all my data is already publicly available, I am not concerned about data leakage. The real concern is that prompt injections get the model to make embarrassing/inappropriate casual claims in relation to the date.

4. Governance Controls

- a. This agent operates only on a static CSV uploaded by the user (I wanted to have it be a live feed to one of the public servers, but that ran too slow for the 5-minute presentation). No persistent state or logging of outputs is maintained unless manually saved by the analyst. This reduces long-term data-retention risk. Relevance AI provides full traceability through each user interaction, which is great for auditability. It also provides me with important summary statistics like runtime and credit/token use. Escalation triggers are built thoroughly and have worked on testing. The escalation trigger explicitly informs the user to contact Benjamin Lee, ensuring human review is integrated into the workflow. If this tool got adopted by my City, I would have access to the paid version which would provide more tracking and governance management properties.

Business Case

The major ROI for this project is deflection and further time saved for data analysts throughout the City. City analysts are in short supply and are often creating and deploying new tools, without time for backtracking and answering questions or simple data queries from past users. Analysts probably spend about 15-20 on each basic query question that they are asked. Whereas the agent takes around 1 minute. Assuming that an analyst receives 5 of these questions a week, they would save 87.5 minutes on average a week, which is about 75 hours a year, per analyst. This is an immensely large amount of time and thus costs. Moreover, this is time that analysts can spend creating new tools and updating old ones to be more contemporary.

This agent can also be used on the City's Open Data Portal to raise the value of the large, raw data files that are currently hosted there. The average citizen does not have the technical skills to pull this level of information from the raw data. The agent can bridge this gap. Error reduction might also be worth considering. It is unlikely that the agent will significantly reduce error rates among analysts, but it may decrease error rates amongst other users.

As far as adoption case is concerned, I think the biggest consideration is that this would democratize data usable through the City. For example, as it currently stands, the only folks willing to use statistics are analysts or technical staff. Whereas this would allow other employees to use the data more effectively and potentially proliferate data literacy throughout the City.

Honest Limits and Next Steps

1. Limitations:

- a. The agent has a hard time clarifying questions with vague prompts. Part of this can be mitigated by explaining users how to interact with the agent, but I will need to make a more significant semantic layer for this agent to be more nuanced.
- b. The agent has, so far, done a good job normalizing the data. But this data has already been generally normalized/cleaned. Implementing additional data will likely be a challenge if it is not rigorously normalized.
- c. The agent does not perform cross-dataset joins or contextual analysis. While this is intentionally limited to preserve correctness and governance, this is a clear limitation of an agent when compared to an analysis.
- d. Performance is suboptimal. I am still troubleshooting why this is, but right now, the agent takes longer than preferred when answering questions. Still faster than emailing/messaging an analyst, but not as fast as a normal LLM.

2. Next Steps:

- a. I need to complete a more sophisticated semantic layer for this data. Including geographic groups, charge families, etc.
- b. Validation checks would be good, but I am not quite sure how to implement these yet. Especially given the slow performance issues.
- c. Connecting to the live dataset on the Open Data Portal is a must eventually. I need to determine how to optimize these connections. When I last tried to connect it, my performance dropped drastically and a simple query took about 10 minutes.